

SIMATS ENGINEERING

ASSESSMENT TOOL – I : SIMULATION TASK

Course Code	ECA1407
Course Name	Embedded Systems
Course Outcome	CO2
Assessment Tool	Simulation Task
Question No.	17
Simulation Task	Develop an Embedded C simulation for an automotive lane detection alert system.
Maximum Marks	30
Bloom's Taxonomy Level	BL3

Evaluation Rubric (30 Marks)

Criteria	Weightage	Level 4	Level 3	Level 2	Level 1	Marks Awarded
Requirements Analysis & Conceptual Understanding	6	☐ 6	☐ 5	☐ 4	☐ 2–3	
System Design & Simulation Model Development	8	☐ 8	☐ 6	☐ 5	☐ 3–4	
Implementation & Tool Usage	6	☐ 6	☐ 5	☐ 4	☐ 2–3	
Testing, Validation & Result Analysis	6	☐ 6	☐ 5	☐ 4	☐ 2–3	
Documentation	4	☐ 4	☐ 3	☐ 2	☐ 1	
Total	30					_____/30

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Output:

Wokwi IDE interface showing the code and simulation of an ESP32-based lane detection system.

Code (Sketch):

```
30 void loop() {
31   if (leftSensor == HIGH && rightSensor == HIGH) {
32     Serial.println("LEFT SENSOR : NORMAL");
33     Serial.println("RIGHT SENSOR : NORMAL");
34     Serial.println("STATUS : VEHICLE CENTERED");
35   }
36   else if (leftSensor == LOW && rightSensor == HIGH) {
37     digitalWrite(LED_PIN, LOW);
38     digitalWrite(BUZZ_PIN, LOW);
39     tone(BUZZ_PIN, 1000);
40     Serial.println("LEFT SENSOR : DETECTED");
41     Serial.println("RIGHT SENSOR : NORMAL");
42     Serial.println("WARNING : LEFT LANE DEVIATION");
43     Serial.println("ACTION : MOVE VEHICLE TO THE RIGHT");
44   }
45   else if (leftSensor == HIGH && rightSensor == LOW) {
46     digitalWrite(LED_PIN, LOW);
47     digitalWrite(BUZZ_PIN, LOW);
48     tone(BUZZ_PIN, 1000);
49     Serial.println("LEFT SENSOR : DETECTED");
50     Serial.println("RIGHT SENSOR : DETECTED");
51     Serial.println("WARNING : RIGHT LANE DEVIATION");
52     Serial.println("ACTION : MOVE VEHICLE TO THE LEFT");
53   }
54   else {
55     digitalWrite(LED_PIN, HIGH);
56     digitalWrite(BUZZ_PIN, HIGH);
57     tone(BUZZ_PIN, 1000);
58     Serial.println("LEFT SENSOR : DETECTED");
59     Serial.println("RIGHT SENSOR : DETECTED");
60     Serial.println("WARNING : BOTH LANES DETECTED");
61     Serial.println("ACTION : CORRECT VEHICLE POSITION");
62   }
63   delay(2000);
64 }
```

Simulation:

Serial Monitor Output:

```
ets Jul 20 2019 12:21:46

reboot (POWERON_RESET),boot:0x13 (SPI_FAST_FLASH_BOOT)
config: 0, SPIW0=0x0
clk_drv=0x0, q_drv=0x0, d_drv=0x0, csh_drv=0x0, h4_drv=0x0, wq_drv=0x0
mode:WIO, clock div:1
load:0x1ffff000, len:1156
load:0x40070000, len:11456
No 0 tail 12 room 4
load:0x40000000, len:2972
entry 0x40000000

=====
AUTOMOTIVE LANE DETECTION ALERT SYSTEM
=====
System started successfully
=====
LEFT SENSOR : NORMAL
RIGHT SENSOR : NORMAL
STATUS : VEHICLE CENTERED
=====
```